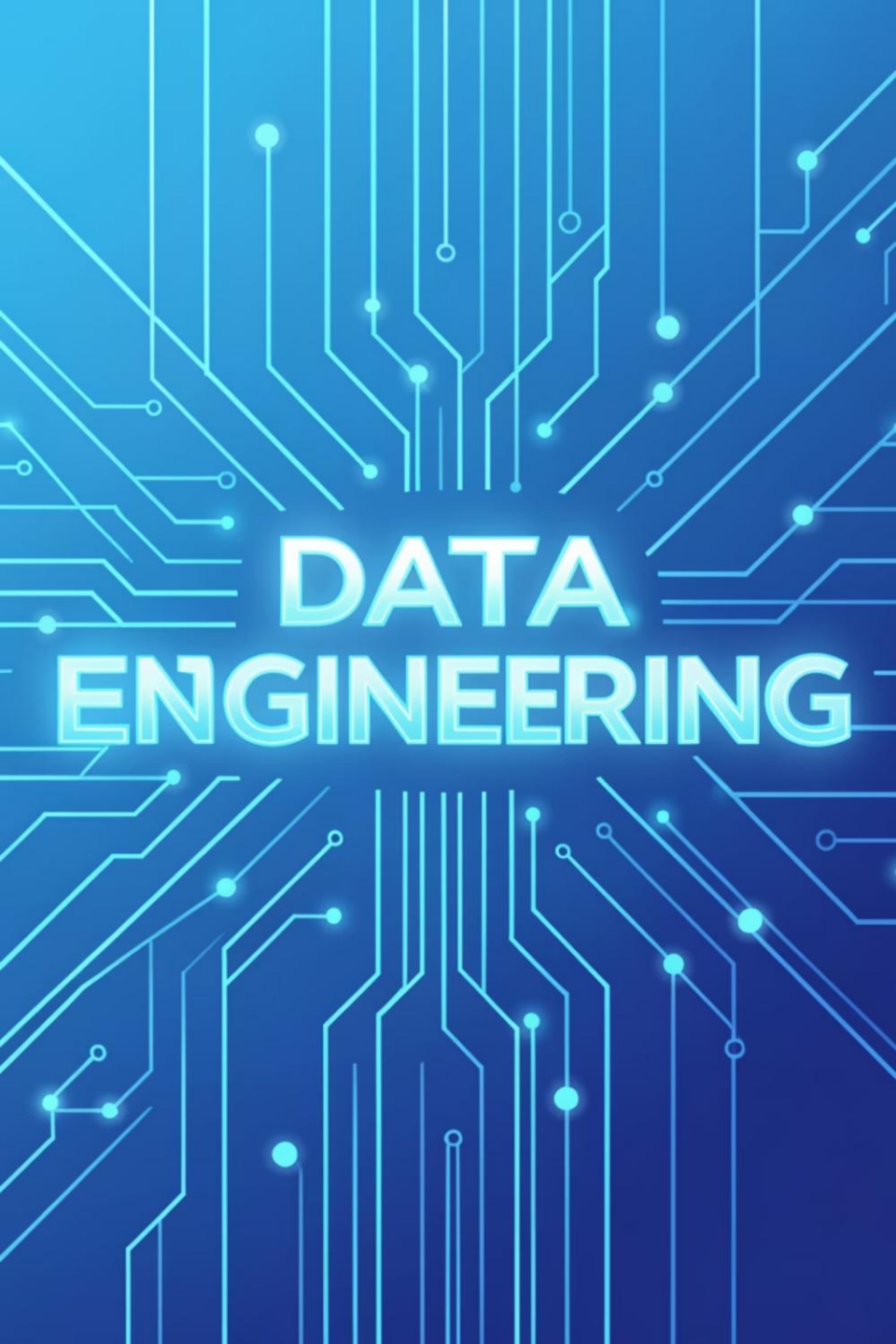




DATA ENGINEERING

Data Engineering Spring '26 – DS463

Apr 02, 2026

Lecture 25

Logistics: Your Course Essentials

Class Timings

Monday (LH6), Tuesday (LH8): 11:30 AM
AM – 12:20 PM

Friday (LH8): 11:00 AM – 11:50 AM

Office Hours

Monday, Tuesday: 10:00 AM – 11:00 AM,
AM, S18 2nd Floor NAB or by
appointment.

Credit Hours

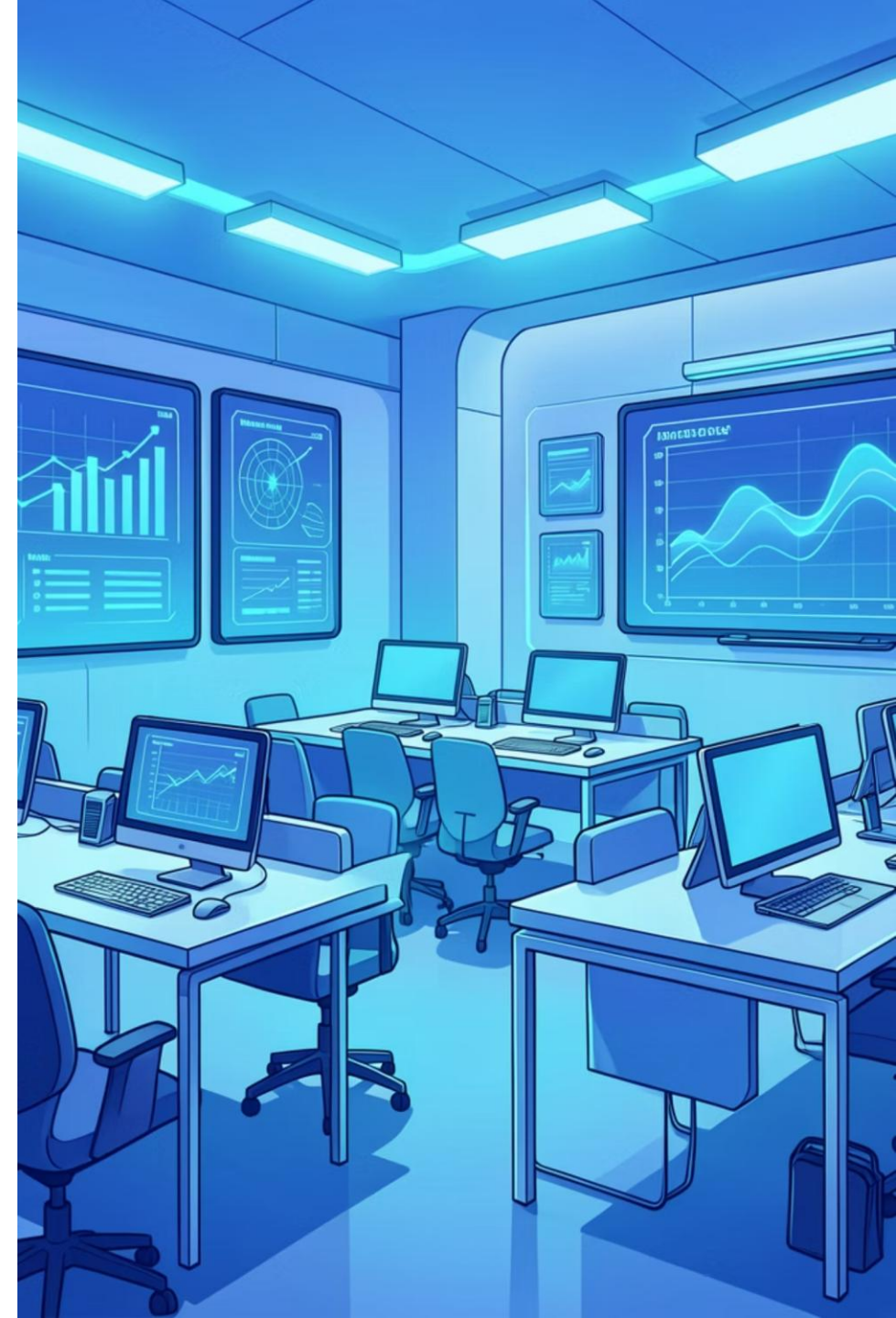
2 hours (Lectures) + 1 hour (Practical
(Practical Session)

Contact & Support

MS Teams: **w5t93yk**

Email: farah.saeed@giki.edu.pk

WhatsApp: Reach out for quick queries



Storage

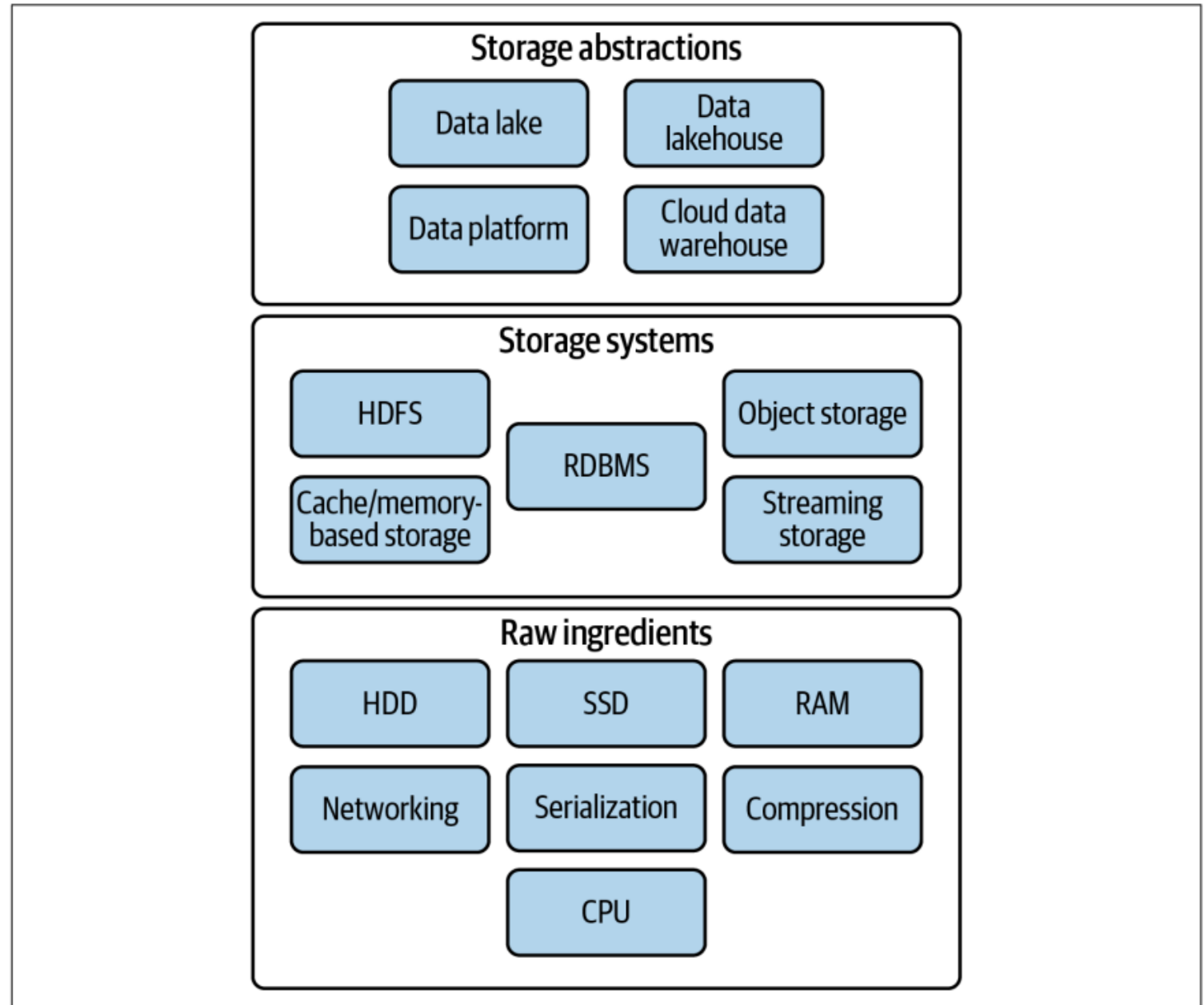
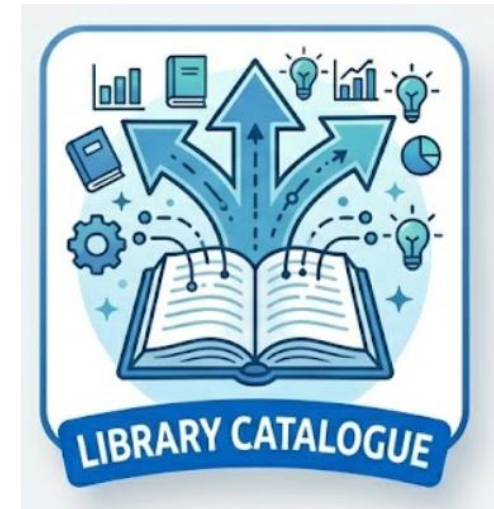


Figure 6-2. Raw ingredients, storage systems, and storage abstractions

Big Ideas and Trends in Storage

Data catalog

- A data catalog is like a library index for all data in an organization.
- It doesn't store the actual data, but stores information about the data (metadata).
- It helps you find, understand, and use data easily.



Big Ideas and Trends in Storage

Data catalog

What does it connect to?

- Connects to many systems like:
 - Databases
 - Data warehouses
 - Data lakes
- Works with both:
 - Operational data (day-to-day systems)
 - Analytics data (used for analysis and reports)

What information does it provide?

- Shows:
 - Where data comes from (data lineage)
 - How datasets are related
 - Descriptions of datasets
- Allows users to edit or improve descriptions

Big Ideas and Trends in Storage

Data catalog

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- Allows users to edit or improve descriptions

Role of a Data Engineer

- Connecting different data systems to the catalog
- Ensuring the catalog stays accurate and updated

Big Ideas and Trends in Storage

Data catalog

Integration with applications

- Ideally, apps directly send metadata to the catalog via APIs
- This keeps the catalog automatically updated

Automated scanning

- The catalog scans systems to collect metadata
- Can detect relationships between table

Big Ideas and Trends in Storage

Data catalog

User interface (data portal)

- Provides a web interface where users can:
 - Search for datasets
 - Explore relationships between data
- Acts like a google search for company data

The screenshot displays the Apache Atlas web interface. On the left, a sidebar contains search filters: 'Basic' (selected) and 'Advanced' (disabled), 'Search By Type' (hive_column), 'Search By Classification' (PII), 'Search By Term' (Search Term), and 'Search By Text' (Search by text). Below these are 'Favorite Searches' (Save, Save As) and a 'PII columns' button. The main area shows search results for '(Type: hive_column) AND (Classification: PII)'. It includes a table with 8 records, columns for Name, Owner, Type, Type, Classifications, Table, and QualifiedName. The table lists various PII columns like providename, emailaddress, ccnumber, nationalid, ssn, and tax_2010 across different tables like prov_view, ww_customers, and tax_2010. The interface also features a 'Columns' dropdown, 'Exclude sub-types', 'Exclude sub-classifications', and 'Show historical entities' options.

Apache Atlas

SEARCH CLASSIFICATION GLOSSARY

Basic Advanced

Search By Type
hive_column

Search By Classification
PII

Search By Term
Search Term

Search By Text
Search by text

Clear Search

Favorite Searches Save Save As

PII columns

Results for: (Type: hive_column) AND (Classification: PII)
If you do not find the entity in search result below then you can [create new entity](#)

Showing 8 records From 1 - 25

☐ Exclude sub-types ☐ Exclude sub-classifications ☐ Show historical entities Columns

☐	Name	Owner	Type	Type	Classifications	Table	QualifiedName
☐	providename	hive	hive_column	string	VENDOR_PII	prov_view	claim.prov_view.providename@cl1
☐	emailaddress	hive	hive_column	string	PII	ww_customers	hortoniabank.ww_customers.emailaddress@cl1
☐	ccnumber	hive	hive_column	string	PII	ww_customers	hortoniabank.ww_customers.ccnumber@cl1
☐	nationalid	hive	hive_column	string	PII	ww_customers	hortoniabank.ww_customers.nationalid@cl1
☐	nationalid	hive	hive_column	string	PII	us_customers	hortoniabank.us_customers.nationalid@cl1
☐	ssn	hive	hive_column	string	FINANCE...	tax_2015	finance.tax_2015.ssn@cl1
☐	providename	hive	hive_column	string	VENDOR...	provider_summary	claim.provider_summary.providename@cl1
☐	ssn	hive	hive_column	string	FINANCE...	tax_2010	finance.tax_2010.ssn@cl1

Page Limit : 25

Big Ideas and Trends in Storage

Data sharing

Giving access to data to others

- Specific data
- Specific permissions

Internal Sharing

- With in organization

External Sharing

- Across organizations

Big Ideas and Trends in Storage

Schema

Answers:

- What format is the data? (table, json, image, etc.)
- What are the data types? (text, number, date)
- How is it connected to other data?

Describes:



Structured

Unstructured

Big Ideas and Trends in Storage

Schema

Two main types

Schema on write

- Schema is defined before storing data
- Data must follow the rules.
- Clean and consistent data
- Easy to analyze later
- Less flexible (hard to store new types of data)

Schema on read

- Data is stored without strict structure
- Schema is applied when reading the data.
- Very flexible
- Can store any type of data
- Harder to analyze later
- Inconsistencies (missing or different fields)

Big Ideas and Trends in Storage

Separation of compute from storage

- Compute (processing) and storage (data) are kept separate.
- Instead of one system doing both, they are handled by different systems.

Older Approach: Colocation

- Low latency
- Example:
Hadoop (HDFS + MapReduce)

Modern Approach: Separation

- Data is stored in one place
(e.g., cloud storage)
- Compute resources are
created only when needed

Big Ideas and Trends in Storage

Separation of compute from storage

- Compute (processing) and storage (data) are kept separate.
- Instead of one system doing both, they are handled by different systems.

Older Approach: Colocation

- Low latency
- Example:
Hadoop (HDFS + MapReduce)

Modern Approach: Separation

- Data is stored in one place (e.g., cloud storage)
- Compute resources are created only when needed.
- Slower but (flexible,scalable)

Big Ideas and Trends in Storage

Separation of compute from storage

Hybrid Systems in Practice

- Combines separation + colocation
1. Caching
 2. Cloud systems

Example: AWS EMR

- Data stored in cloud storage
- Temporary cluster created to process data
- Intermediate data stored locally
- Final result saved back to storage

Example: Apache Spark

- Loads data into memory (RAM)
- Processes data very quickly
- Releases resources after job

Big Ideas and Trends in Storage

Data Storage Lifecycle and Retention

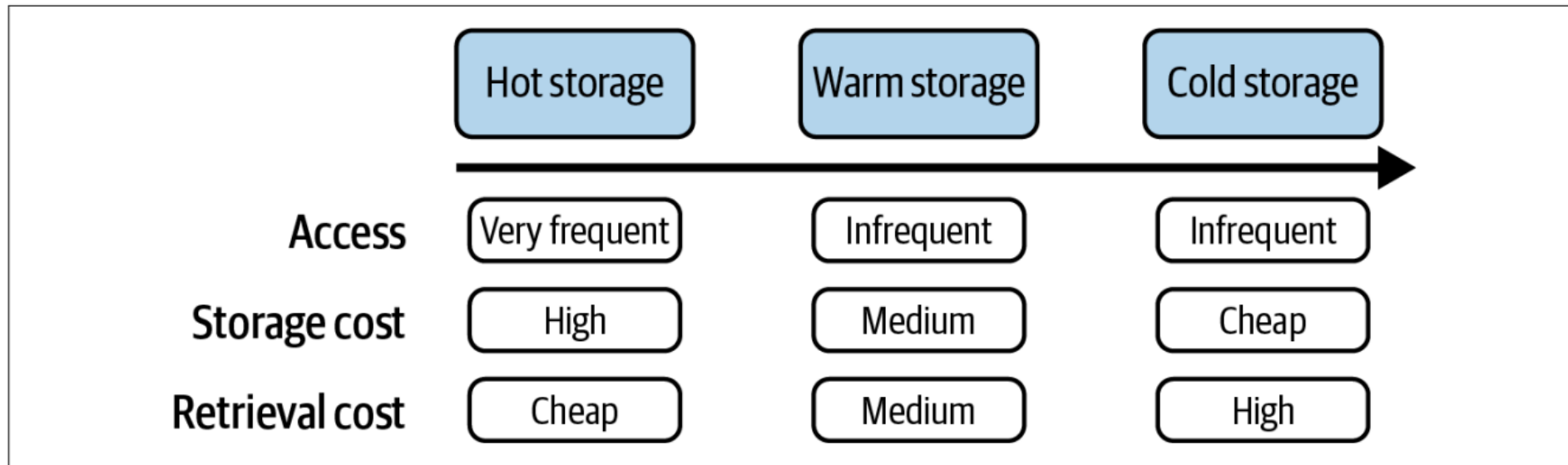


Figure 6-9. Hot, warm, and cold data costs associated with access frequency

Undercurrents

Security	• Principle of least privilege
Data Management	• Data Catalogues • Metadata management
Data Ops	• Observing • Monitoring
Data Architecture	• Reliable, Scalable, Elastic, Durable
Orchestration	Storage allows data to flow through pipelines, and orchestration is the pump.
Software Engineering	Storage through Infrastructure through code

Storage – Data Storage Systems

Data Lakehouse

Data Lakehouse combines:

- Data lake (raw storage)
- Data warehouse (structure & management)
- Stores data in object storage (like a data lake)
- Adds data warehouse features for better management

Features

- **Structured tables & schemas**

Support for Updates, deletes, merges

- **Versioning (history & rollback)**

Can go back to older versions of data



- Not all data needs structure
 - Some can stay **raw/unstructured**
 - Some can be **organized into tables**
- Use structure **only when needed**

Storage – Data Storage Systems

Data Platform

A **data platform** is ecosystem for integrated tools:

“All-in-one system” for:

- Storing data
- Processing data
- Analyzing data

Example:

BigQuery = **Storage + SQL + ML + Visualization** in same platform

Storage – Data Storage Systems

Stream to batch storage architecture

Combines

- Real-time (streaming) processing
- Batch (long-term) storage

Data is collected as a stream

Then sent to multiple systems (consumers)

Real-Time Consumers

Process data instantly

Generate:

- Live statistics
- Dashboards

Batch Storage Consumer

Stores data for

- Long-term use
- Batch analysis

Airflow DAG

